The Role of Touch in Memorization: The Differences Between Digital and Paper Mediums

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Abstract

The COVID-19 pandemic has changed the educational sphere forever. Since then there has been a significant shift in modality with instructors preferring digital mediums. This study aimed to examine if there is a relationship between digital or paper mediums and memorization ability. We randomly assigned a sample of 60 undergraduate students to either a paper or digital condition. The participants were asked to memorize a list of 12 non-valenced words on either medium. They had 45 seconds for the memorization task and were then given a word search. Researchers then administered a short, easy word search for participants. The researchers then asked participants to recite as many words as they could. Results indicated that the medium did not have a significant impact on memorization rather the number of interactions did. Participants who interacted more with either medium tended to receive higher scores. These findings have important implications for education and teaching models. Our research emphasizes the importance of meaningful interactions with educational materials.

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The COVID-19 pandemic has brought about significant changes in the world. In the academic sector, instructors were forced to adopt an online modem for teaching. Even as classes have slowly transitioned back to an in-person modality, instructors are still utilizing many electronic practices since the pandemic. The reliance on digital technology has become abundantly clear. However, while online learning and electronic documents have streamlined communication and reduced paper waste, they may not be helpful for memorization.

Many psychology studies have explored the possibility of differences in mediums on memorization. Porion (2016) tested whether a paper or digital medium affected memorization. The group of researchers hypothesized that digital/computer screens were more effective due to the screen being higher definition. The researchers recruited 72 French children in their third and fourth years of secondary school, with an equal number of boys and girls and an average age of 13.9 years. All participants had experience using computers, were native French speakers, and had normal or corrected-to-normal vision. The study aimed to assess comprehension of an explanatory text about the bloodstream, which was constructed with a hierarchical structure and analyzed using various metrics, including the Lix formula. The Lix formula is used to calculate the readability of a text based on the number of words, the number of sentences, and the number of long words in the text (Porion et al., 2016). The text was considered challenging, with a score of 52.4 on the Lix formula. The researchers had the children read a text and answer questions falling under different question types (semantic, surface, inferential). After gathering data and analyzing the results, the researchers found no significant effect between medium and memorization. This study is crucial because determining the best medium for learning can revolutionize education and psychology. While no significant difference was found between

mediums, if future studies show that digital mediums are less effective than paper, it could drastically alter our approach to education and psychological research.

As current undergraduate students, we found our interest in this topic crucial to our current education model. Additionally, compared to the Porion (2016) study, digital mediums are being used more than ever due to COVID-19 and technological advancements. The drastic increase in usage of digital technology may lead to different results. Our aim was to determine which platform would be more effective for memorization and which platform instructors and students should use moving forward. Additionally, we sought to gain further knowledge on how to conduct future psychological studies.

We decided to replicate the Porion (2016) study with a conceptual replication type. The purpose of replicating this study was to answer the question: which medium, digital or paper, is best for memorization? We wanted to test the same theoretical idea however, we wanted to account for our demographic as well as focus on short-term memory. While the Porion (2016) study was thoroughly conducted and provided great insight into how different types of questions played a role in memorization, the study mainly focused on reading comprehension as opposed to short-term memorization. While reading comprehension is a category of memorization, it can be more complicated than a memory recall task. The text likely contained valenced words, making the paragraph more memorable as the reader attributes positive or negative emotions and memories alongside the paragraph. In addition, we did not have access to the same demographic as we are undergraduate psychology students. Furthermore, reading comprehension may have affected the results of the studies due to the reading comprehension level of the children as they were all in 3rd or 4th grade; it doesn't attest to the advancement of their individual reading comprehension skills. We addressed these limitations by studying college students primarily

from the University of San Francisco, ensuring they have all received a similar amount of education and are more proficient in reading. Furthermore, we changed the test to focus on memorizing non-valenced words, so there was no positive or negative association with the words aiding in the memorization process. Their measurements may have also impacted their results, and adding the interaction measurement could clarify their results.

Our team hypothesized that a paper would be a more effective medium for memorization. We suggested that paper has tangible qualities that electronics do not. Marshall (2017) suggested participants may be more inclined to touch the paper by pointing or holding it and thus, enhance their learning ability. Our team decided to conduct a short-term memorization task (a different test compared to the original study). We added an interaction factor with the medium to see if that affected learning and memorization. We hypothesized that these changes would lead us to a definitive result. Due to the interaction factor, we contradicted the hypothesis from Prion (2016) and hypothesized that paper mediums are more effective for learning and memorization due to the tangibility aspect. By studying patterns of interaction with a given medium students can better optimize their learning experience.

Method

Participants

The study population consisted of a convenience sample of college students. The participants were required to meet the following inclusion criteria: be at or over the age of 18 and be enrolled in a university. The sample size in this study came to N = 60. Of the 60 participants, 30 were assigned to the digital medium group and 30 were assigned to the paper medium group. Participants were randomly assigned to each group.

The average age of the participants was 20.5 years old (*SD* = 1.5), ranging from 18 to 25 years. The gender distribution of the participants was 53.3% women, 40% male, and 6.7% nonbinary. Additionally, 26.7% of the participants reported their ethnicity as White, 28.3% as Asian, 11.7% as Black, 1.7% as Pacific Islander, 16.7% as Hispanic/Latinx, 10% as mixed, and 5% as Other.

Procedure

The present study utilized a between-subjects design method to examine the potential impact of medium type (ie. digital or paper) on memory retention. Participants were randomly assigned to either the digital or paper medium group.

Upon arrival, participants were given a list of 12 non-valenced words and instructed to memorize the terms for a period of 45 seconds (Megalakaki et al., 2019). Proceeding the memorization period, participants were administered a short word search task, which was selected from X and categorized as "Kindergarten-level" (refer to Figure 1). This task was designed to increase the external validity of the experiment by introducing a mediator that served as a potential distraction, mimicking real-world conditions. Researchers gave the participants unlimited time to complete the word search.

Figure 1

After completion of the word search task, participants were then asked to verbally recall the memorized words. The use of a verbal recall task is a widely accepted and effective method of memory performance (Gavett, B. E., & Horwitz, J. E., 2012). Researchers gave the participants unlimited time to complete the recall task.

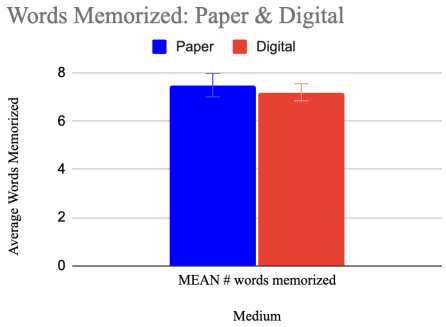
Measures

The dependent variable in this experiment was the number of words memorized which was measured at the end of the study. Additionally, we measured the number of interactions someone had with the medium. This served the purpose of supporting our hypothesis. We defined one interaction as any form of physical touch with either medium (ie. pointing to words on a paper, scrolling on a computer screen).

Results

We conducted two independent samples t-tests to compare the scores of the two conditions, paper and digital. The results showed that the paper condition had a slightly higher mean of words memorized ($M = 7.5$, $SD = 2.688$) and a higher interaction score ($M = 2.33$) compared to the digital condition, in which participants memorized around 7 words ($M = 7.2$, $SD = 1.94$) and had only one interaction ($M = 1.2$) Although there was a slight difference, an unpaired t-test we found that the p-value was 0.62, indicating there was no significant difference in memorization between the two mediums. Therefore, we fail to reject the null hypothesis, which suggests that medium has no significant difference in memorization.

Figure 2



Although we did not find a statistically significant difference between medium and words memorized, we did discover a correlation between words memorized and the number of interactions. We used the Pearson correlation coefficient to determine the strength of the relationship for each medium. The digital medium had a higher correlation coefficient ($r = 0.20$) than the paper medium ($r = 0.09$), indicating interaction may be slightly more important for digital mediums. We combined both of the datasets and found a composite correlation coefficient of $r = 0.13$. These scores indicate a weak positive correlation between the number of interactions and words memorized, regardless of the medium.

Figure 3

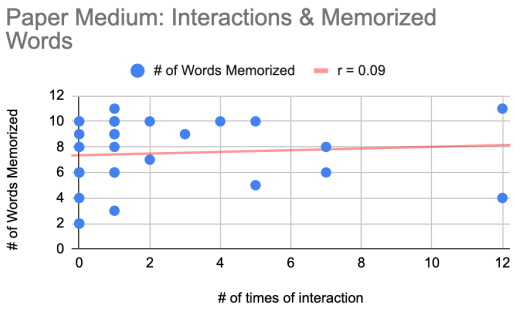


Figure 4

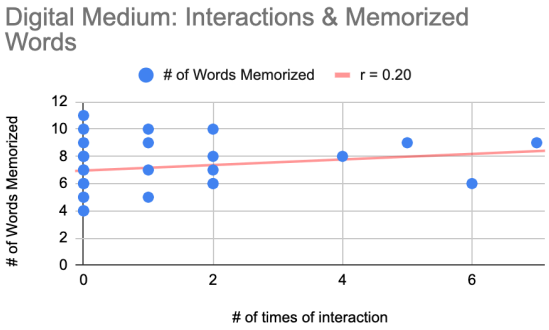
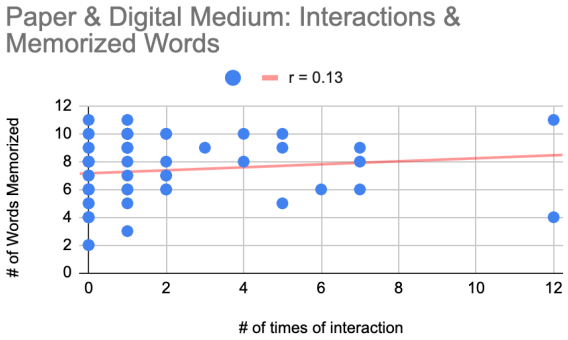


Figure 5



Discussion

Sometimes students and educators have strong preferences for using paper or digital mediums. This contention motivated our exploration of whether mediums made a difference in memorization. We found no significant difference in memorization for either medium. Our findings run parallel to the Prorian (2016) study. However, the additional measurement of interactions did help us find differences in memorization ability. Generally, participants did slightly better when they interacted with their medium.

Limitations

The present research has two limitations that are worth mentioning.

First, we recruited through a convenience sample. One of our researchers (Rosales) gathered most of their participants through their choir practice group. Choir practice requires skills involving memorization and verbal recall. That particular group of participants may have had a slight advantage because of their prior background. Therefore, the demographic of participants may have influenced the results. This may have contributed to a biased sample. Thus, our findings should be generalized with some caution.

Second, the procedure was conducted by three different researchers. Each participant was given the memorization task at different times and in different environments. We observed that

environmental conditions varied considerably. We observed background noises such as music, laughter, and talking during some of the tasks. The external noise may have distracted participants and thus, influenced their performance on the memorization task. In the same vein, time was varied. In addition, the researcher (Hayworth) conducted the memorization task at 11:30 pm. The same participant reported they were "really tired" before one of our researchers conducted the experiment. Cognitive functions are often weakened when paired with sleep deprivation or exhaustion (Alhola, P., & Polo-Kantola, P., 2007). Well-rested individuals may have had an advantage and received a higher performance score. While not a large number of participants reported feeling tired and/or they participated at night, these conditions may have affected the accuracy of some of our results.

Implications

The debate on whether paper or digital mediums are better for memorization has been a highly contended issue, particularly in the era of digital technology. Our findings, as well as the Porion (2016) findings, support that both mediums are equally effective for memorization. Thus, we can infer that the transition from paper to digital modality in our schools due to COVID-19 has not had a negative impact on students' ability to memorize.

Our findings have practical implications for educators and students. Our findings suggest that educators who have a preference for paper mediums may not necessarily need to enforce that policy. Rather, students should be given a choice to choose between either paper or digital that best suits their learning preferences. In addition, our study highlights the importance of interacting with their choice of medium. Educators may want to adapt or modify their teaching strategies so that students have active engagement with their material. Furthermore, students can adapt interactive techniques into their study routines to optimize their learning experience. The

intersection between technology and education is forever growing. Thus, this study offers support for digital technology to be used in the classroom.

As for future research topics and methods, exploring different modes of interaction could provide more insight into our findings. For instance, do different tactics such as holding a piece of paper or pointing to words on a piece of paper have a difference in memory skills? Additionally, there is the potential to explore combining paper and digital mediums. Many students and educators use a combination of both mediums in the classroom, Hybrid mediums may hold different results for memorization than just paper or digital. In our study, we assessed short-term memory. Future researchers could replicate the study and change the duration between the memorization task and the recall task. By conducting further research on this topic educators and students are able to make better decisions regarding educational and study routines.

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